You can use Telemetry events to write to StatsD reporters such as Datadog, or you can write your own custom handlers. Plus, due to standardization, you can usually integrate a new library's Telemetry events quickly into your existing handlers.

Phoenix provides an in-depth guide[5] on how to use Telemetry events to monitor your application that uses Phoenix. Many other libraries include similar documentation on their telemetry events.

Telemetry is just one type of observability. Another big one is backend performance monitoring. Most of the major players like New Relic and Scout APM support Elixir and provide guides on how to integrate it. This also applies for bug-tracking platforms like Bugsnag, Rollbar, and Sentry.

The most important thing about observability is to do it early. You don't want to have a production problem and not be aware of it or not be able to diagnose the cause of it.

Let's change gears and look at how LiveView is powering the future of real-time apps in Elixir.

Real-Time Apps with LiveView

Elixir has a solid foundation in real-time application development. It started with Phoenix Channels, which I wrote all about in *Real-Time Phoenix [Bus20]*. But there's been an awesome library that has completely dominated real-time application development in Elixir—LiveView. (It's so popular that it's often the first thing people hear about Elixir.)

LiveView[6] is a library that enables a new breed of real-time applications. It is server-centric, which means that all of the page HTML is processed on the server and efficiently sent to clients on update. So, if a change happens on the server, connected clients are instantly and automatically updated.

The magic of LiveView is that it enables you to create powerful applications with minimal JavaScript. All of the page interactions are processed by the LiveView process, so the server stays in control of the page interactions. In my experience, the amount of JavaScript needed to create a single-page app experience is reduced by 95% or more!

LiveView was briefly mentioned in Chapter 7, Serving Requests with Phoenix, on page 117, but it's so prevalent in the Elixir ecosystem that it deserves its

5. https://hexdocs.pm/phoenix/telemetry.html
6. https://hexdocs.pm/phoenix_live_view/welcome.html